



SAN DIEGO STATE
UNIVERSITY

College Of Engineering
Department of Mechanical Engineering
Final Report



ME 491 Engineering Design-01: Senior Design

Project 20, Team MPM7-20-B, Ember Generator 2.0

Team Ignite

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Date Submitted: May 8th, 2026

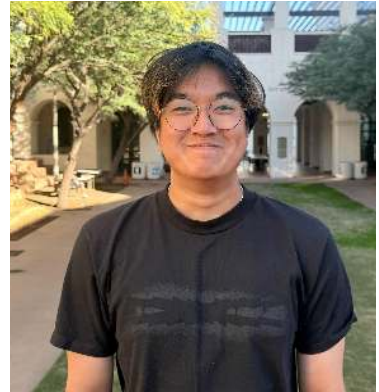
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Table of Contents:

ABSTRACT.....	6
1.0 Project Overview.....	7
2.0 Project Management.....	9
2.1 Final Organization Chart.....	9
2.2 Final Schedule.....	9
2.3 Updated Risk Management Cube.....	12
2.3.1 Risk and Issue Management.....	13
2.4 Final budget.....	15
3.0 Final Design.....	16
3.1 System Level Diagram.....	18
4.0 Bill of Materials.....	19
5.0 Manufacturing and Assembly.....	20
5.1 Manufacturing Plan.....	20
5.2 Assembly Plan.....	22
5.3 Quality.....	22
6.0 Requirements Verification and Final System Demonstration.....	24
7.0 Knowledge Management.....	29
8.0 Summary.....	30
9.0 References.....	31
Appendix.....	32

Table of Figures:

Figure 1: Team Organization Chart.....	9
Figure 2. Project Milestones Gantt Chart.....	10
Figure 3. Spring Gantt Chart.....	11
Figure 4: Risk Cube.....	13
Figure 5. Infrared of Combustion chamber in use.....	14
Figure 6. Charcoal ember Test.....	14
Figure 7. Wooden dowel ember test.....	14
Figure 8: As Built Ember generator.....	16
Figure 9. Top down View of Combustion Chamber.....	17
Figure 10. System Level Diagram.....	18
Figure 12. Tools Used in EIS-106B.....	21
Figure 14. Wind Tunnel Subsystem mock run with various paper.....	25
Figure 15. Embers being blown out of Wind Tunnel hitting trays at 3 and 5 feet marks.....	25
Figure 16. Wind Sensor Rev P Testing.....	26
Figure 17. Battery power shut off.....	27
Figure 18. Emergency shut off for Massage Gun.....	27
Figure A1. Wind Speed Test Results.....	33
Figure A2. PCB Schematic for Massage Gun.....	36
Figure A3. Projected Schematic for Circuit.....	36
Figure A4. Electronic Circuit for Massage Gun.....	36

Table of Tables:

Table 1: Risk and Mitigation Steps Table.....	12
Table 2: Bill of Materials.....	19
Table 3: Requirements and Verification.....	24
Table 4. Verification of Requirements.....	28

ABSTRACT

Wildfires have become more common in California in recent years and it has begun to affect Wildland-Urban Interface (WUI) areas. To help support research on ember behavior and its effects in WUI zones, the Ember Generator 2.0 Project focuses on designing a portable system that is capable of creating and projecting out embers in controlled conditions. We designed a stainless steel combustion chamber with an electrical shaking mesh mechanism positioned above a straight pipe wind tunnel which enables consistent ember production with adjustable wind speeds. We conducted engineering analysis to confirm design requirements, including heat transfer calculations for material selections and windflow analysis to determine ember exit velocity and trajectory using projectile motion calculations. We also conducted testing at the Cal Fire Facility to make sure all requirements were met and the generator was ready for lab and on field testing for wildfire research. The design satisfies the requirements of portability, safety, quality, and performance and will support ember ignition research in WUI areas.

1.0 Project Overview

Wildfires are an increasingly severe issue, posing significant threats to natural ecosystems and human communities especially in regions like California. Although not all wildfires are harmful, low-intensity fires are actually critical in many conifer forests as their destructive potential increases dramatically under the influence of these fires. Under windy conditions, the problem is further intensified as firebrands, also known as embers, are carried by the wind and spread fires further and igniting vegetation, structures and other flammable materials. This is major in structure loss in Wildland-Urban Interface areas (WUI), yet there is limited research on how embers ignite vegetation and materials around buildings. The Ember Generator 2.0 project focuses on designing and creating a portable ember generator to support wildfire research at San Diego State University. The device is meant to help researchers and other engineers understand firebrand behavior, how far they travel, how the wind extends their reach, and how they ignite nearby fuel under controlled conditions. To further understand the impact of firebrands, the Ember Generator 2.0 project was created in order to further test and understand the causes of wildfires in order to help with prevention and better ways to stop occurring ones. The goal of this project is to create an ember generator that simulates wildfire conditions by designing, building, and testing a system that can accurately replicate ember generation under controlled conditions and on a smaller scale. Requirements for this project include

- Combustion and ember generation: the system must be capable of burning wood pellets to generate consistent embers for at least 10 minutes.
- Wind simulation: include a wind mechanism that can vary wind speeds up to 5 m/s.
- Ember flight distance: embers should travel a distance of approximately 3 meters from the exit nozzle to model ember spread.
- Portability: system must be lightweight, portable, and easy to assemble/disassemble to accommodate for testing.
- Safety: implement fire safety measures like fire extinguishers, fire blankets and protective equipment (PPE). The project also has a set of key deliverables that must be met which include

The project will require background information on key engineering tools and require learning how to create:

1. Preliminary design sketches and system specifications
2. Detailed CAD drawings of the system
3. Bill of materials (BOM) with cost estimates
4. Functional prototype of the ember generator
5. Demonstration of the system's capabilities, including ember generation and wind speed control
6. Test plan outlining the procedures for validating system performance
7. Test results and analysis, including any modifications made based on testing
8. Comprehensive report detailing the design process, challenges faced, and solutions implemented
9. Recommendations for future improvements and potential applications

Dr. Luca Carmignani is supporting this project with the help of the wildfire research lab at San Diego State University. He specializes in fire science and hopes to create this generator as a way to study firebrand behavior and apply it to understand wildfires and fire behavior in wildland urban interfaces (WUI). He is our fire advisor and with his background we are able to create this generator in hopes of being able to create an impact in wildfire prevention. The goal is to help Dr. Carmignani understand the role of embers in

wildfire destruction in WUI environments and aid in the prevention of wildfires. Listed below is Dr. Carmignani's contact information.

- Name: Luca Carmignani
- Phone/Fax: (619)594-3599
- Primary Email: lcarmignani2@sdsu.edu

The Ember Generator 2.0 team plans on working with Dr. Carmignani to help accomplish the project goals. The group consists of five senior engineering students to design, build and test a working ember generator. Listed below are everyone's respective roles and contact information.

- Krystal Morera : Team Lead | kmorera1829@sdsu.edu | (619) 549-0305
- Tyson Chavez : Design Lead | tchavez4931@sdsu.edu | (760) 470-3444
- Ronin Toy: Manufacturing Lead | rtoy8753@sdsu.edu | (818) 661-9767
- David Gray: Quality Lead | dgray1635@sdsu.edu | (619) 653-8099
- Mike Khuu : Electrical Lead | mkhuu5430@sdsu.edu | (858) 265-9301

Each team member is responsible for their own defined roles and responsibilities. The project requires collaboration and independent time to create a successful project. Project meetings are conducted during scheduled class time on Mondays at 2:00pm and by need basis with additional meetings as needed based on workload and progress. To ensure the project is progressing with minimal disruption, the team is under sponsor guidance. The team lead is responsible for weekly progress reports due every Friday and weekly sponsor meetings to review progress, address questions, and to discuss any issues. Working hours are located in facilities like the EIS-106 building, machine shop, Love Library, and Dr. Carmignani's Lab (E331) for design, fabrication, and assembly. Off-campus testing will be conducted at the Cal Fire facility in El Cajon with necessary permits and supervision of Dr. Carmignani to test our prototypes.

The project has a budget of \$3000 that will cover the duration of the 9 months of project time. This will be a design and material intensive project. The budget will go towards material procurement and fabrication. The complexity of the generator requires application of engineering design and analysis skills to ensure a functional and reliable final product. The following highlights the key skills essential to create this system.

Key Skills

- Solidworks/CAD design
- Arduino/Electronics
- Machining

General Design Knowledge

- Fluid Mechanics
- Thermodynamics/Heat Transfer
- MATLAB/Excel
- Design Fabrication of combustion chamber and high-temperature enclosures
- Sensors (thermocouples, motors)
- 3D Printing
- Welding
- Material Science
- Safety Features and Protocol
- Environmental Testing

2.0 Project Management

This semester was largely focused on the procurement and manufacturing of the design that our team came up with in the fall, as well as tests to ensure all the requirements were met. There were several setbacks in both procurement and manufacturing, but those were expected risks that were successfully mitigated. By the end of the semester we were able to manufacture and assemble the ember generator with time to conduct two on site full assembly tests, as well as proving to meet all the set design goals in the system description document. The original design changed this semester due to having to navigate some fabrication issues but the final design will be shown below. The original design the team came up with can be found in detail within the Fall Final Report.

2.1 Final Organization Chart

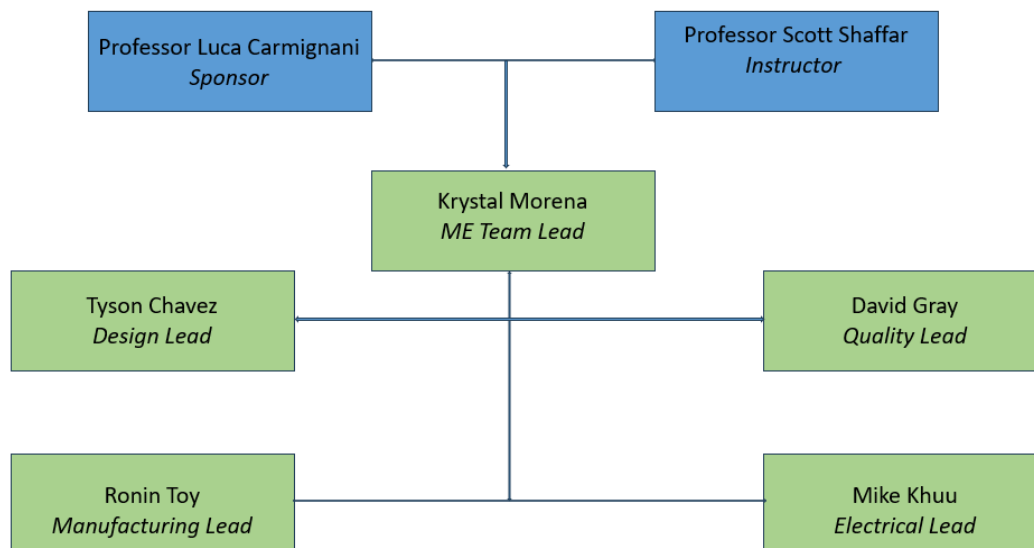


Figure 1: Team Organization Chart

This semester our team organization chart stayed relatively the same except this semester some team roles were shifted. We still had guidance from our sponsor Dr. Luca Carmignani and reported to Dr. Shaffar any updates, deadlines, and weekly status reports. In figure 1, the graphic shows the breakdown in roles and responsibilities for each member and who we would report to.

2.2 Final Schedule

TeamGantt was very helpful for keeping the project on track in the spring. It provided a clear view of how much time was left, and how much time had already been spent on the project. Using it to organize and rearrange various deadlines was essential to keeping the pace of the project organized and on track.

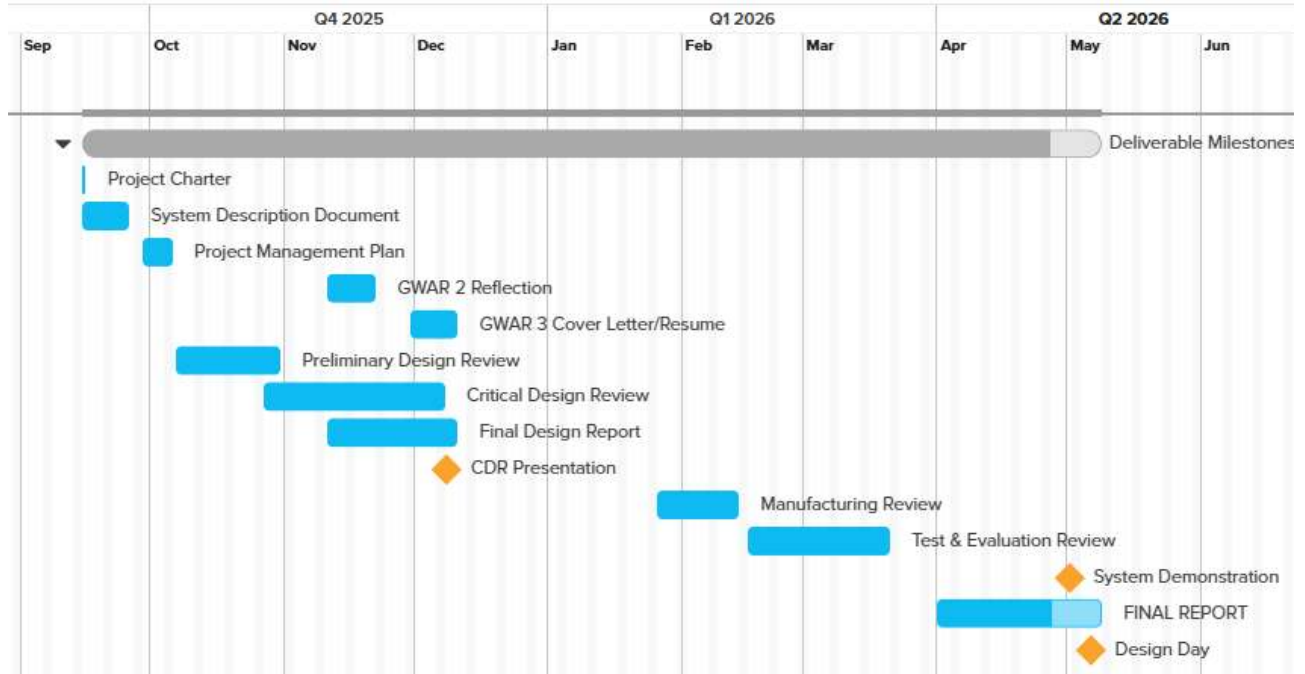


Figure 2. Project Milestones Gantt Chart

Pictured above are our project milestones for the whole year in TeamGantt. This has remained unchanged, we have continued to keep up with the assignments and presentations on time.

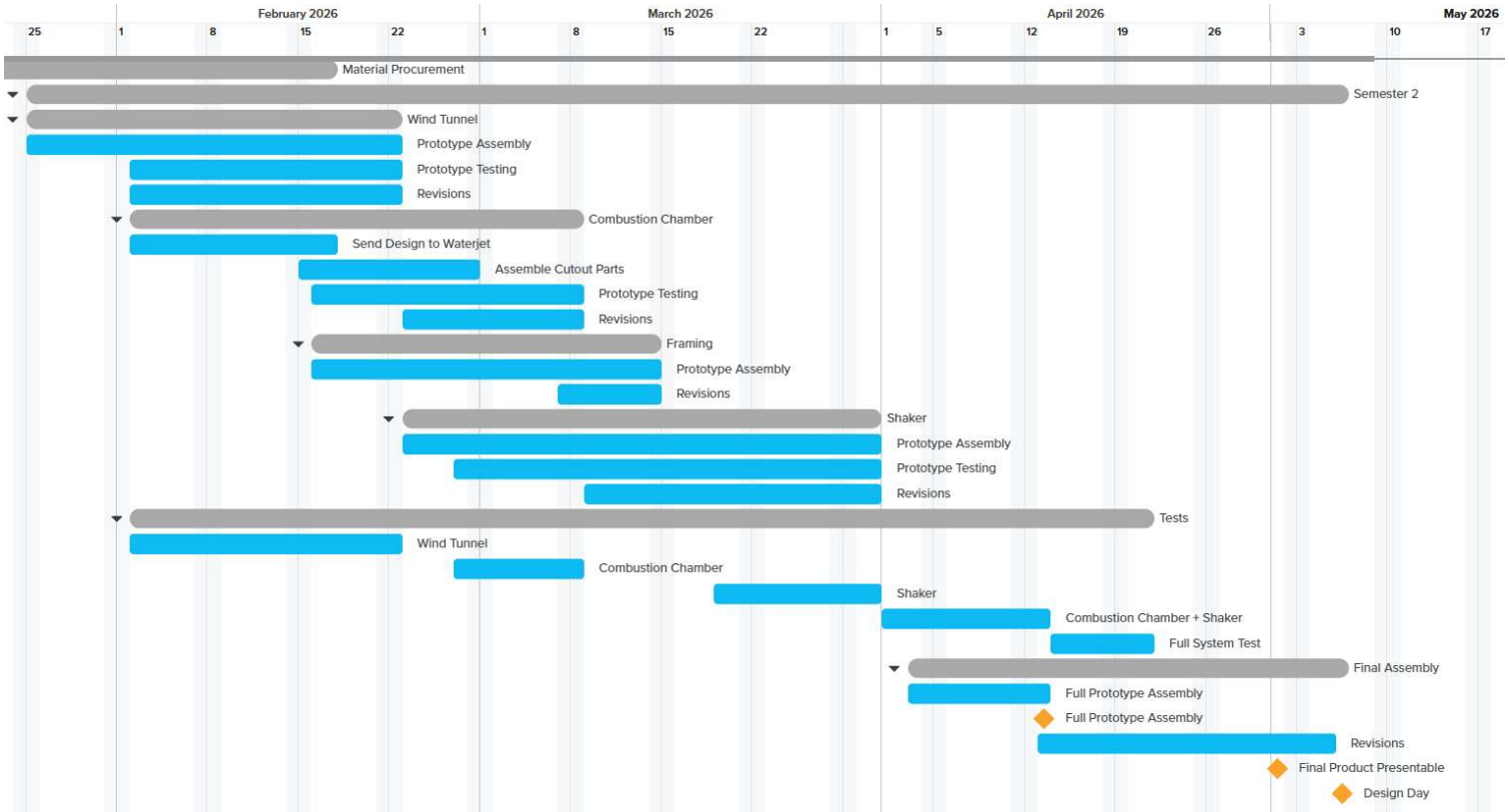


Figure 3. Spring Gantt Chart

Figure 3 shows our schedule that was followed by the team for the spring semester. When compared to the tentative plan we had for spring in the fall semester, it is clear we did not know at the time how long some of the tasks would take to complete. Before, we had planned to have the full prototype built by March, when in reality it was only finished on April 13th in time for the on site test on the 14th. This was due to several setbacks both in procurement and manufacturing.

At the top of the chart in figure 3, it is visible that the material procurement went as far into the second semester as the 17th of February, when it was planned to be finished over winter break. There was a miscommunication with the department, so what we thought had been ordered was actually not ordered over break and needed to be reordered. The waterjet in SDSU was also down, so we needed to outsource our sheet metal cuts. This took an extra two weeks to research and arrange. There was also some difficulty welding the stainless steel that we needed for its heat resistance. The shop in E-105 only had MIG welding, where TIG welding was required for our material. While there was TIG welding elsewhere on campus with the Baja Racing team, they were also incredibly busy at the time so we opted to outsource the welding as well.

There was a slight miscommunication with the welders, so our combustion chamber was made differently than initially expected. This required a near complete redesign of the shaker, which is why that component in the schedule took by far the longest to complete. Thankfully since we had initially set only a loose deadline of early March and had ordered extra materials, there was plenty of wiggle room to work with solving this problem. Ultimately the solution we came up with required several more water jetted parts, but by that point the waterjet in E-105 was back up and running again.

Once the shaker redesign was complete the full prototype test was scheduled, and went very smoothly when compared to the manufacturing. The rest of our time after the tests were spent fixing minor issues found while testing and working on the required presentations and reports.

2.3 Updated Risk Management Cube

The project has multiple technical risks that could happen and affect our generator if not prepared for and carefully mitigated. The risks range from physical damage to our generator, damage to our environment and to the people involved, and not achieving the intended outcome from the finalized product and realizing our goals. The table below goes over eight different risks that are listed by priority and matched with a level of consequence, level of likelihood, and the mitigation method we plan to enforce to prevent the risk from happening or at least lower the likelihood from happening.

Table 1: Risk and Mitigation Steps Table

Priority	Risk Statement	Level of consequence	Level of likelihood	Mitigation Method
1	If the flames get too intense, then the system can become too hot and cause structural failures and warpage.	5	2	To mitigate this, we will install a quick switch to be able to turn off actuator instigating the embers, and furthermore will have a button to fully turn off all the electronics to the system to allow it to cool down.
2	If the specific material/fuel doesn't produce enough/robust embers, then it would prove difficult to blow the embers out and be able to ignite the desired target.	4	1	To mitigate this, we have different types of fuels and sizes to test to find which ones produce the most ideal embers.
3	If prototype is not built by march, then there won't be enough time to test and change any necessary parts	4	1	To mitigate this, we will have to make sure we follow a strict timetable and be ready to order before end of semester is over
4	If the initial ignition fails or falters, then the generator will fail to be able to produce embers.	3	2	To mitigate this, we will do extensive testing with the desired ignition component with multiple prior prototypes and extensively check that the wiring to the ignition source is connected properly and correctly.
5	If there is a delay in material delivery, then the materials won't be here in time, and we won't have sufficient time to build our generator.	4	1	To mitigate this, we will have found multiple back up suppliers with the same/similar products that we need in order to mitigate the risk of not being able to receive the desired materials in a timely manner.
6	If there is clogging in the combustion chamber then the ember production process will be brought to a halt, thus bringing the whole system to a halt.	3	1	To mitigate this, we have incorporated a shaking device to help sift the embers through the mesh in the combustion chamber.
7	If material sourcing becomes scarce, then the project schedule will be pushed, and project cost may increase.	2	1	To mitigate this, we will have multiple options ranging from quality to quantity to ensure we have various options.
8	If the ember generator isn't at the desired height, then the generator might not be able to efficiently hit the desired target at the minimal distance.	2	1	To mitigate this, we can add adjustable legs or have extra stilts to prop up the generator and have done analysis of air flow to make sure embers project to the desired range.

LIKELIHOOD	5					
	4					
	3					
	2			4	1	
	1		7,8	6	2,3,5	
		1	2	3	4	5
CONSEQUENCE						

Figure 4: Risk Cube

The figure above shows the risk cube that contains all the risks discussed. Each risk is represented by a number that is based on when they were presented. The mitigation methods are meant to shift the consequences and likelihood into a green or yellow box in order to make the project safer, achievable, and meet all the necessary requirements and deadlines required for the project. After analysis and further designing we were able to lower the likelihood of all risks and even eliminate potential risks. Using heat transfer calculation, windflow to determine ember trajectory calculations, and prototype testing, we have successfully lowered the likelihood of all the risks.

2.3.1 Risk and Issue Management

This section of the report will be dedicated to further explanation of each of the risks and how each of the risks in the risk cube was managed to a lower level of likelihood compared to the cube in the critical design review. The risks were numbered in order of how important they were, 1 being the most critical risk. The first risk was deemed the most significant risk because the fire built in the combustion chamber has a chance to get out of hand and if not handled properly could result in damage to the machine and spillage into the surrounding environment. What the team did throughout its fabrication to mitigate this was to control and limit the amount of fuel being burnt at a time. This was done by creating a makeshift trap door to open and close to release the fuel onto the shaking tray. With this, the amount of burning fuel that was released to the shaking tray was limited so that the fire could never grow too large. However, after fully testing the machine, the heat was always contained in the combustion chamber and would only go out little through the bolts. An image of this can be seen below in figure 5. This trap door can be seen below in figure 9 in the final design section 3.0.



Figure 5. Infrared of Combustion chamber in use

Risk number 2 was tantamount to the success of our project. Being the ember generator, if our machine did not provide enough embers then the project would have essentially failed. To circumvent this issue, the team did extensive testing on various amounts of fuel to see what would create the most and viable embers. This testing, along with the implementation of the shaking mesh greatly improved the number of embers and their quality, lowering the likelihood of the risk greatly. Figures 6 and 7 below show how we tested the materials in our fuel composition.



Figure 6. Charcoal ember Test



Figure 7. Wooden dowel ember test

Risk number 3 was a big issue because of the difficulties of acquiring permits to test was a well known issue among the group. As a result, the team aimed to finish as soon as possible in order to obtain these permits early to test and troubleshoot issues. Adhering as best as we could to the strict scheduling, we were able to overcome some delays with manufacturing and build an ember generator well in time for two separate testing sessions, lowering the likelihood of this risk.

Risk number 4 ties into risk number two, where failure to properly ignite the fuel would inherently lead to the production of little to no embers. This issue was resolved by introducing the makeshift trapdoor mentioned previously which kept all the fuel near the hot rod long enough to catch. With that implementation into the design, the likelihood of this risk was greatly reduced.

Risk number 5 is similar to the issues in risk number three which threatens the project as a whole by delaying the time we would have to complete it. This problem was solved by ordering as soon as possible over the winter break in order to have the materials to work with when we came back. Furthermore, the team ordered an excess of some parts to prepare for the worst and have parts on standby to troubleshoot any mistakes that we would encounter.

Risk number 6 would affect the efficiency of the project. If the machine were to ever clog it would be inside the combustion chamber and due to the nature of the project, the combustion chamber would be too hot while running it to easily and safely unclog it. To resolve this issue, the trapdoor helps limit how much fuel is released onto the shaking tray and the amount of fuel for each batch is a low amount. Having such a control over the amount of fuel being put into the machine each time greatly reduces the risk of the machine clogging. There is also an opening at the back side of the shaking tray to help remove some of the fuel for after testing or if deemed necessary.

Risk number 7 was an issue the team deemed early on when creating a bill of materials to order the parts. Most of the steel, aluminum framing, and other nuts and bolts were easy to acquire due to how normal they are to the industry. However, an issue we faced was a certain type of massage gun or a similar acting actuator that was not battery powered. The options were very limited for this part and we ended up needing to settle with another option that had a battery. This option however was fixed by rewiring the massage gun to remove the battery and attach it directly to a DC power source. This not only made finding an actuator easier, we were also able to choose an effective option for a relatively lower price.

Finally, risk number 8 imposed a problem to one of the requirements of the project to shoot out embers at a desired range of 3-5 feet. This problem however was directly solved by making the legs on the ember generator adjustable. With this solution, the ember generator can be preset to a desired height. The fans' speed was also powerful enough to blow the embers well in and over the range at its default setting. This design implementation can be seen below in the bottom half of the picture of figure 8.

2.4 Final budget

The final cost of all of the parts that were actually used in the final assembly came out to about \$1735.27 according to table 2, this does not include manufacturing costs or miscellaneous spare parts that were donated by the Fire Lab at SDSU. For a more detailed breakdown of the total costs of the project, look to section 4.0 Bill of Materials. Ultimately we spent more than was initially expected, but still came in under the budget of \$3,000.

3.0 Final Design

Our final As-Built design was extremely similar appearance wise to what we had planned in the fall semester. Upon closer inspection, there were some major changes specifically concerning the shaker tray and combustion chamber.



Figure 8: As Built Ember generator

Compared to the design from the previous semester, there was only one main change. The frame, air tunnel, and fan subassemblies remained unchanged. The plan from last semester was to have the shaker tray directly below the combustion chamber. As a result of poor communication, the combustion chamber was not fabricated to the correct shape, forcing a design change. Our group decided to combine the shaker tray and the combustion chamber by using the cut out intended for the separation tray for the shaker instead. While this did solve our immediate issue, we still needed a way to separate the already burnt, ember generating material from the unburnt material. We had to create a new part, a swinging separation flap, in

order to allow the hot rod to ignite material. As seen in figure 9, the separation flap worked perfectly, and allowed us to contain the shaker tray completely within the combustion chamber.



Figure 9. Top down View of Combustion Chamber

This was our solution to the most glaring problem with the design, and in the end it was successful. Our team was able to combine two of the most critical components into one, allowing for the heat to be contained better, the design to be more compact, and have less moving pieces. While the combustion chamber mishap was not planned, it ended up having a positive effect on the design, making it more compact, less prone to issues, and safer in regards to heat. The rest of the design remains completely unchanged, and more details about the air tunnel, frame, and fan can be found in the final design review.

3.1 System Level Diagram

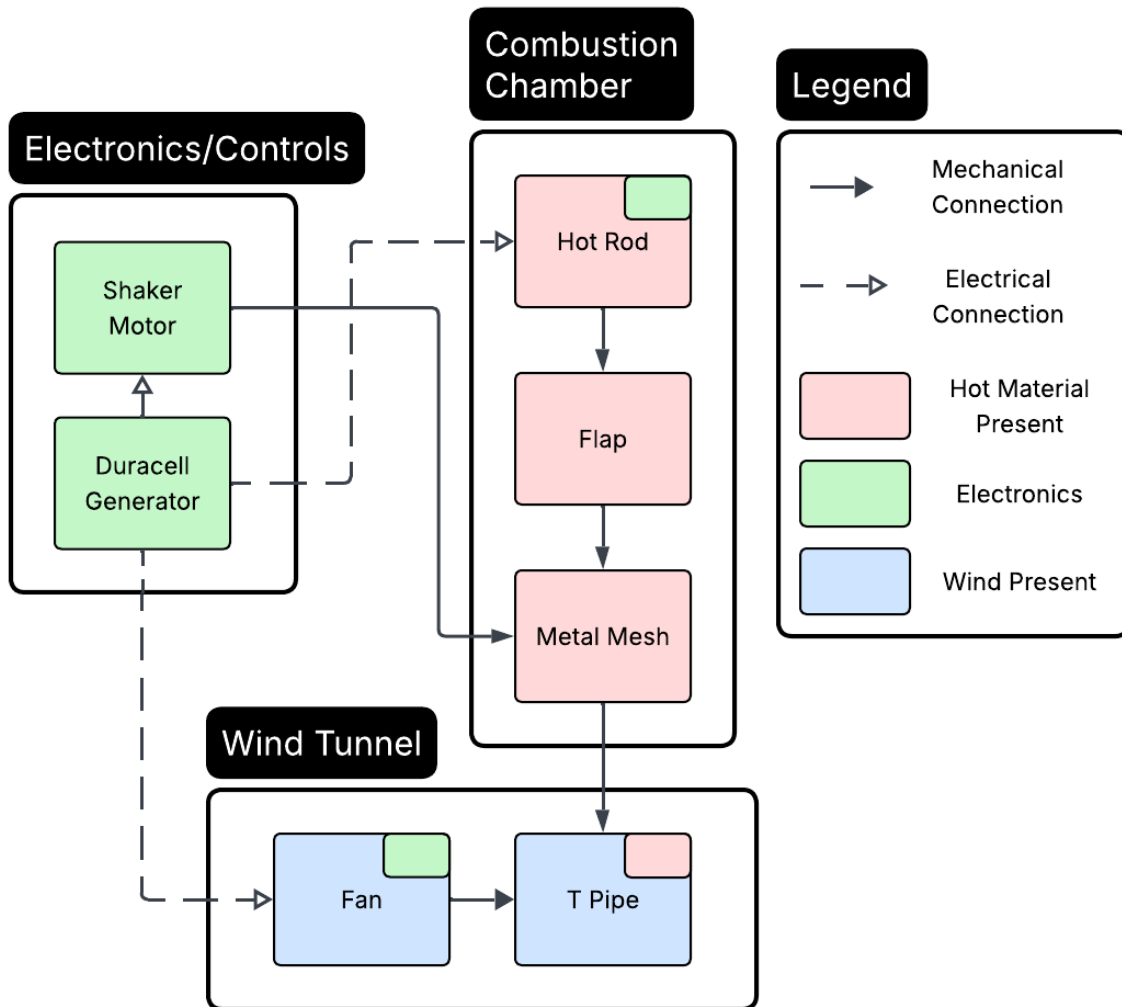


Figure 10. System Level Diagram

The system level diagram remains largely unchanged from the final design report. The overarching design is still the same, but some redundancy has been removed and one element has been swapped for something else.

The redundancy was having two straight pipes in the wind tunnel, when one of the pipes would have only had wind in it. Now, every section in the wind tunnel serves a purpose. The replacement is changing from a sliding separator to a trapdoor flap. Once again, this is simply removing unneeded complexity from the design. The previous design included a spring hinge on a metal flap that could be pulled to open. Now it is simply a pivoting metal sheet, as was explained in section 2.0.

The electronics system was also simplified somewhat, as during testing it proved much more practical to measure the temperature and wind speed externally as opposed to within the device itself.

4.0 Bill of Materials

Table 2: Bill of Materials

Subsystem	Part	Supplier	Price	Quantity
Combustion Chamber	S414 2x4 Stainless Steel Sheet	Metals Depot	\$193.96	1
	Steel Mesh Overall Thick 0.051" Opening: width 1" length 0.3" Strand :width 0.099" thick 0.051"	McMaster-Carr	\$26.78	1
	Massage Gun	Walmart	\$40	1
	M4 Screws	Home Depot	\$1.75	12
	M4 Locknuts	Home Depot	\$1.75	12
	200 watts / 120 volts, 24" wire lead hot rod	Amazon	\$25.28	1
Wind Chamber	Inline Duct Fan, 6-Inch 400 CFM with Variable Speed Controller	VEVOR	\$54.99	1
	2013K17 Galvanized Steel Straight Reducer for Standard Duct Duct Size 6 to 4, 6" Female ID x 3-15/16" Male OD	McMaster-Carr	\$21.40	2
	T-Pipe	McMaster-Carr	\$44.88	1
	3/4 in. x 10 ft. White PEX-B Pipe	Home Depot	\$8.87	1
System Framing Supports	Aluminum Bolt Together Framing (10ft)	MVMaster-Carr	\$78.61	1
	Straight Framing Bracket 8809t61	McMaster-Carr	\$11.44	4
	Bracket Fastener Sets 1 1/2"	McMaster-Carr	\$4.59	5
	Aluminum Telescoping Rails(1 1/2"X 1 1/2")(4ft)	McMaster-Carr	\$91.64	4
	Aluminum Telescoping Rails(1 3/4"X 1 3/4")(2ft)	McMaster-Carr	\$71.82	4
Electrical Control System	Portable Duracell 500W Generator	Duracell	\$399	1
	Arduino Board and Wires	SDSU Bookstore	\$76.95	1

Table 7 shows the bill of materials for all the components, divided into their respective subsystems, chosen to be used in the final design. This table includes the name of the part, the supplier chosen for the part, the price per part, and the quantity of each part needed to be ordered. Certain components such as the generator, lead hot rod, and Arduino board set were provided by the SDSU fire lab at no cost but their costs were still provided on the table.

The total cost of the parts used for the ember generator, not including the costs of shipping and tax, came out to be \$2,280.48. The starting budget of the project is \$3,000, of which \$2722.11 was spent, which leaves the team with \$277.89, about 9.26% of the budget. The team did its best to procure cost effective, yet reliable and quality components. While trying to make cost effective decisions, the team did face a dilemma while choosing a ready made actuator. Ultimately, it was decided that it was best to use a massage gun in replacement for an actuator to save on costs.

When choosing reputable vendors, the team wanted to keep the amount of vendors to a minimum in order to simplify shipping and to maximize compatibility of parts. McMaster-Carr was chosen as the primary supplier for a multitude of reasons. One being the fact that McMaster-Carr had a very vast inventory for the team to find any needed components. Also, they were known to be very fast and reliable with shipping. A feature of McMaster-Carr that also helped with the design side of the project, was the fact that they provide spec sheets and CAD models of all their parts. This made it easier to be able to integrate the exact parts into our CAD assembly. The hardest part to obtain was the massage gun, since we wanted a wired one rather than one powered by a battery, which might create a risk of blowing up near heat. Unfortunately, the wired one we found sold out so we decided to get a battery powered one and just modify it to be powered directly by the generator rather than a battery.

Originally, the team had planned for 45% of the budget to be saved for unforeseen replacement of broken parts, re-designs, and extra parts when production began in the spring. This proved to be a smart decision, because not only were more parts needed to be ordered due to redesign, but there were unforeseen manufacturing costs. At first, the plan for manufacturing the combustion chamber was to be done in house at the school machine shop, but the school's water-jet broke down and the machine shop didn't have the right welding tools for stainless steel. This caused the team to allocate some of the budget to outsourcing the manufacturing to outside contractors. Overall, this added on a total of \$441.63 to the expenses of the project.

All in all, the biggest lesson learned from part procurement was how important planning for the worst was. Two ways the team tried to mitigate future financial and scheduling issues were to order double the amount of stock sheet metal and to leave around half the original budget in the original plan. Both these strategies turned out to be very crucial to the completion of the project. Ordering double the amount of sheet metal allowed us to fix any mistakes with the extra stock without having to wait for more to be shipped, especially since this part's shipping time was the longest. As described previously, having allotted around half the budget for unforeseen problems and redesigns, allowed the project to stay under budget even when having to contract out some of the manufacturing process.

5.0 Manufacturing and Assembly

5.1 Manufacturing Plan

Our initial manufacturing plan was to make everything on campus, however that had to change once it was clear that the waterjet was not going to be fixed in time for our parts to get made. Most of our parts, such as the wind tunnel and framing, were off the shelf and therefore required little to no manufacturing on our part. The plan was to assemble the simpler parts of the assembly while simultaneously manufacturing the combustion chamber and shaker tray.

This went somewhat according to plan as can be seen in Figure 2, where the wind tunnel was assembled early while the combustion chamber parts were sent off to get waterjetted. Once the cut parts came back we were also able to start working on the framing in parallel with the bending and welding needed for the combustion chamber. The bolt together framing needed to be cut to size before it could be bolted together.



Figure 11. Sheet Metal Bender in Machine Shop

The bending was done in the machine shop with their sheet metal bender. It was a very manual process, and the springback of the material made it very difficult to judge the correct amount of bending. We did some research on how much springback should be expected, and even made a reference of the angles needed out of paper, but it was still very difficult to get right. Once the bending was done it was very clear that the parts did not make a close enough fit to be welded. This led to a week of grinding and hammering the parts into shape to get it to the welders in a state that was weldable. To do this we used the grinder and tools in EIS-106B.



Figure 12. Tools Used in EIS-106B

Once the welding came back it was clear that the shaker needed a redesign. Using some of our spare sheet metal we sent our design off to Tu to waterjet. Once waterjetted, we used the drill press in EIS-106B to match the bolts and rivets we needed. The combustion chamber itself was also drilled into with the same drill press so that it could be bolted into the framing. Once that was complete all that was left was to assemble all the parts.

5.2 Assembly Plan

As stated previously the assembly plan was largely running in parallel to the manufacturing plan. Most of our parts, such as the wind tunnel, fan, and framing were all off the shelf parts designed to fit together already, so those were simple enough to assemble. Of note in the wind tunnel were the wind aligning pipes, those were put in with a simple friction fit. While one pipe did come loose during testing, this was fixed during later repairs and none of the other pipes have shown signs of failure.



Figure 13. Pipes in Wind Tunnel and Shaker Tray in Combustion Chamber

The framing was built around the wind tunnel, and there were very few issues with bolting together the framing. Once the holes were drilled into the combustion chamber those could be bolted into the framing as well. The slider was slightly more difficult as the full slider tray does not slide out of the slot it was built in. This was done by first only riveting one side of the slider tray together, then angling the rails so that they fit the gap before widening them and bolting together on the other side.

For the electrical assembly, there was some difficulty with attaching the massage gun to the framing, as the vibration would loosen most attempts at holding it steady. Ultimately we decided to wrap it in insulating material in order to dampen the vibrations on the bolts while also holding the massage gun steady.

5.3 Quality

As a whole, the fabrication process for our generator was largely free of defects. With almost every part that was manufactured, there was minimal effort required to make it fit into the design. When we built the combustion chamber, there was some grinding that was needed in order for the corners to fit correctly for welding. We believe this was due to operator error when folding the sheet metal. While this would have been beneficial to avoid, it did not hurt the project significantly.

Another quality issue we ran into was the burrs formed as a result of cutting our framing. For every aluminum framing piece we cut with the bandsaw, there was burring left on the aluminum. This was not a major issue, as a few seconds with the file would fix the issue, but it was another step we had not anticipated. In the end, it was not a big enough issue to try and avoid it. The most efficient thing for the team to do was to cut and file, ensuring the parts would meet our quality standards and prevent people cutting themselves on the frame.

The biggest issue we encountered was not necessarily a quality issue. It was a miscommunication between the welding shop we used to fabricate our combustion chamber and our team. While we do not consider this a quality issue, it did require some necessary changes to our design.

6.0 Requirements Verification and Final System Demonstration

Table 3: Requirements and Verification

#	Requirement	Verification
R1 - 001	The ember distance should range from 3-5 ft	TEST/ANALYSIS -Run the system with a target 3-5ft away and make sure a sufficient amount of embers hit the desired target
R1 - 002	It shall include a mechanism to control and vary wind speeds up to 16 ft/s	TEST/INSPECTION -Run the fan separately to make sure the ordered part performs as desired. Run it again attached to the system with a wind speed sensor to make sure the wind generation continues efficiently through the whole pipe.
R1 - 003	The system must be capable of burning wood pellets to generate embers	TEST -Run the combustion chamber separately to make sure the hot rod is efficiently igniting the fuel and observe the ember generation
R1 - 004	The system should be able to generate embers consistently for 10 minutes	DEMONSTRATE -During testing, run the system for at least 10 minutes and observe the ember generation.
R1 - 005	Safety features to prevent overheating and ensure safe operation	INSPECTION -Have easily accessible off switches/buttons for all electronic parts. Have multiple thermistor in different areas of the apparatus to monitor the temperature. Have a detailed safety procedure for operating the system.
R1 - 006	Portable enough to fit in a van or back of a SUV as well as be carried by two people	DEMONSTRATE -Assemble and disassemble while testing. Calculate the weight to make sure it can be carried by 2-3 people(50 lbs per person) over varying terrain. Have 2-3 group members carry it during testing..
R3 - 001	Permission to test the burning on site.	INSPECTION -Get approved and receive appropriate permits for testing at desired sites

The project has 7 main requirements that need to be verified in order to classify the ember generator as functioning as desired. Table 3 is referencing requirements outlined in the SDD with their respective requirement numbers.

First, the system must be able to project embers at least 1 yard away from the apparatus, which will be verified through test and analysis. The test will consist of setting up a desired target 1 yard away from the apparatus, running the system, and observing if the embers reach the desired target. The analysis will be on wind flow based on blower airflow, ember properties, and entrainment behavior inside the wind tunnel to calculate the distance the embers will travel. — This requirement was verified with a mock run of just the wind tunnel subsystem. This mockrun consisted of just the fan being on and putting a variety of different papers through the opening at the top of the t-pipe. These papers consisted of ripped up notebook paper, tissues, and some gift wrapping paper. The result of this mock run confirmed that the system could blow materials of a similar weight well over 5 feet. Furthermore, when we tested the full system at the Cal fire Facility, the actual embers and other burning fuel shot well over the 5 feet tray.



Figure 14. Wind Tunnel Subsystem mock run with various paper

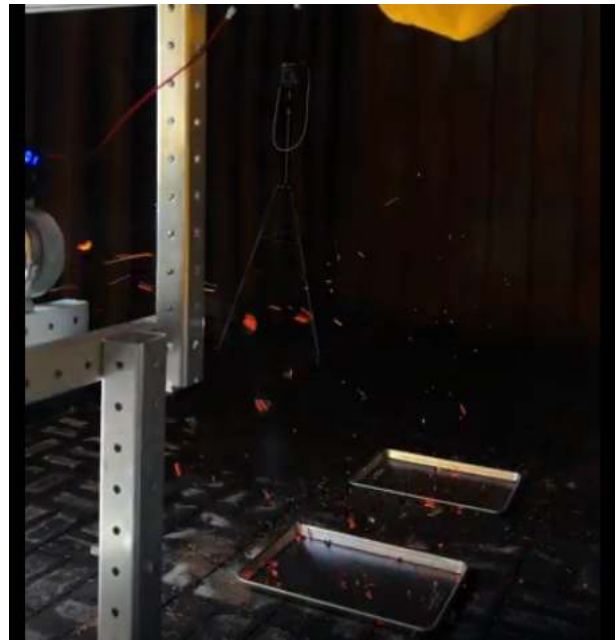


Figure 15. Embers being blown out of Wind Tunnel hitting trays at 3 and 5 feet marks

Second, the system needs a mechanism to control and vary wind speeds up to 16 ft/s, which will be verified by test and inspection. An inspection is first needed to make sure that the ordered Inline Duct Fan, 6-Inch 400 CFM with Variable Speed Controller performs properly, and then a test is needed to make sure it performs properly when integrated into the system. — This requirement was verified with a wind sensor, namely the Wind Sensor Rev P. The wind sensor was set up through an arduino and monitored on a laptop. When testing, the wind sensor was held up to the end of the tunnel while the fan was tested from low to high speed. Figure 16. below will show how the data was conducted and collected. The data from these tests were collected on a computer and can be seen in the bar graph A1 in the appendix.

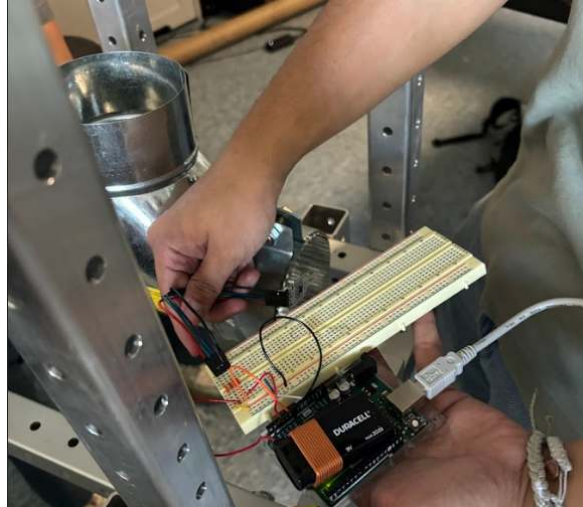


Figure 16. Wind Sensor Rev P Testing

Third, the system must be able to burn wood dowels/coconut coals primarily with some other fuel choices to generate embers. To verify this, the team plans on testing the combustion chamber separately to make sure a sufficient amount of embers are produced. Fourth, the system must be able to generate embers consistently for at least 10 minutes, which the team plans on verifying by running the system and demonstrating that it is able to complete this task. — This requirement was verified by a full system test when the team went out to the Cal fire Facility. During the testing, it was seen that our machine properly generated embers and provided a consistent stream of embers. The coconut coals provided with the consistent stream of embers and the wooden dowel fuel acted as larger flaming chunks. The system is set up so that while the generator is making embers, another batch can be loaded and prepared to fire more kinds of embers.

Fifth, there must be safety features implemented to prevent overheating and safe operation. The team plans on inspecting the shut off switches and thermistors throughout the system to make sure they function properly before running the system. — This requirement was verified during testing at the Cal fire facility via the thermal camera. The team decided on a combustion chamber of stainless steel so that it could contain most of the heat. Since the combustion chamber subsystem is mostly separate from the rest of the system and has such a high heat capacity, most of the heat lies in the combustion chamber through conduction of the burning fuel. Some of the heat travels outwards to the structure through the bolt that keeps it in place, but it does not travel further than that. The radiation also does not go too far so that the rest of the chamber is relatively safe to touch with the proper PPE. There is an emergency shut off switch for the massage gun on the platform hosting the electronics. The final emergency shut off switch would then be on the battery and would cut all electricity to all the components of the ember generator. These shut off buttons can be seen below in figure 17 and 18.



Figure 17. Battery power shut off



Figure 18. Emergency shut off for Massage Gun

Sixth, the apparatus needs to fit in a van or back of a SUV as well as be carried by two people. It is planned to verify this requirement by two methods, demonstration and analysis. It was found that generally 50lbs is the safe weight one person should be able to carry, and after weight analysis it was calculated that our system would weigh about 20 lbs. To demonstrate that the system meets this requirement, the team plans on transporting via car and setting it up. — This requirement was verified by a demonstration of putting on and taking off the legs which can be done by at least two people. The ember generator was also demonstrated to be carried by two people into a car and to certain test sites proving that it is portable and able to fit into a standard sized car. Due to the design, the ember generator without legs was less than 50 lbs, reaching well below the OSHA standards of 50 lbs per person.

Seventh, the team must apply for and get approved for testing in certain wilderness areas and the Cal Fire Facility in El Cajon. This will be verified by inspecting the permission documents to make sure the team is allowed to do testing in the designated locations. So far, a preliminary test has been done for requirement R1-003, by testing the prototype combustion chamber to see if embers were generated and the prototype succeeded in doing so. — This requirement was verified by our sponsor who reached out to the Cal fire facility and got us the official permission to test on their grounds. With their supervision and test site, the team was able to complete all the other verifications and tests needed to properly test the ember generator.

These requirements were verified throughout the manufacturing process during each manufacturing milestone. A table of the tests conducted to verify each of the verifications will be shown below in table 9. Further testing to verify these requirements were then officially done at the Cal fire testing facilities in which all the requirements as mentioned above were officially met. Each of these tests were done with the relative proper equipment. These equipment and instruments were an anemometer (the wind sensor rev p), a thermal camera (to test/show for hotspots), tape measurer for the distance, a stopwatch for timing, and multiple cameras for documentation. The materials used to conduct the tests were wooden dowels, coconut coals, wood chips, and a power generator. The safety equipment used was fire resistant gloves, glasses and masks for the smoke, and since the tests were conducted at a Cal fire facility, a water hose was provided in the case of an emergency. The testing was conducted mainly at the Cal fire facility and some dry runs were conducted at San Diego State University campus.

Table 4. Verification of Requirements

Requirement	Test / Verification method	Objective	Result	Date Completed
R1 - 001	Dry Wind Test	Put some shredded notebook/tissue paper to imitate embers through the wind tunnel and record the distance	This results of this test were that the tissue paper went the farthest at up to 9 feet and the notebook paper landed around 6-6.5 feet at the fans highest setting	2 / 23 at the fire research lab at SDSU
R1 - 001	Full System Test	Running the system fully, use the proposed fuel composition of wooden dowels and coconut charcoals through prior testing and record the distance that the embers go	The results of this test were that the burnt dowels reached the 3 feet, and 5 feet tray. The embers from the coals also reached those marks and went further	4 /14 at the El Cajon CAL FIRE facility
R1 - 002	Dry Wind Test	Use the anemometer to see if the system would reach a required 16 ft/s	The data from the anemometer showed the max setting on the fan reaching a speed of 20.69 ft/s	3 / 23 at the fire research lab at SDSU
R1 - 003	Full System Test	The ember generation is best tested when the full system is being tested as they all work together to create the most amount of embers from burning in the combustion chamber to falling on to the shaking tray	The results of this test were that the wind tunnel was able to shoot out a lot of embers that came from the burning fuel from the combustion chamber which was further agitated on the shaking tray	4 /14 at the El Cajon CAL FIRE facility
R1 - 004	Full System Test	Running the system fully for 10 minutes with all the electronic components on and seeing if the system is capable of consistently generating embers	The results of this test was that the system safely ran as intended for over 10 minutes while producing a consistent stream of embers	4 /14 at the El Cajon CAL FIRE facility

Table 4 Continued

Requirement	Test / Verification method	Objective	Result	Date Completed
R1 - 005	Electronics Systems Test	A dry run of just turning on the individual electronic components to see if the emergency shut off button on the battery and the one connected to the massage gun works properly.	The results of this test was that each of the components completely stopped when the emergency shut off button on the battery and the one for the massage gun was pressed	3 / 23 at the fire research lab at SDSU
R1 - 006	Demonstration	The demonstration will consist of attaching and detaching the legs. While the legs are detached the machine will be transported to either a vehicle or a testing site. This transportation and adding/removing the legs should be achievable by two people	The results of this test are that the legs are easily put on and taken off by two people. The machine is light enough to move into a car and around to any test site.	4 / 14 at SDSU when getting it ready to transport to the Cal Fire Facility
R3 - 001	Demonstration	A testing permit request will be sent to the Cal Fire Facility to test at their site	The permits were approved and the team was allowed to go onto the facility on 4 / 14 and 4 / 21 for a couple of hours each testing day	4 / 6 was when the permit was approved for the testing dates of 4/14 and 4/21

7.0 Knowledge Management

Throughout the past two semesters, our group has learned many valuable lessons in multiple different areas related to this project. There were things that went smoothly, things that did not, and things we would change. We learned lessons concerning communication, planning, and scheduling.

Two of the biggest problems we had to deal with had to do with lapses in communication. To begin with, our combustion chamber was not welded the intended way because when our metal pieces were dropped off, the final design was explained to the welders. They were not given any drawings or a CAD model to look at, which ultimately led them to weld the chamber incorrectly. If we could go back, we would make sure to drop off the sheet metal with a drawing so that we get the correct result. The other lesson we learned concerning communication had to do with the procurement of material. Before the winter recess, our group was under the impression that our materials had been ordered. When we returned we

found that only a couple things had been ordered. This set us back a few weeks, but was not detrimental to the project timeline.

One problem that we were able to avoid completely was due to our planning. When ordering material, we ordered double the amount of sheet metal needed in order to cover any unexpected rework. This ended up being extremely helpful, as we had plenty of extra sheet metal for other parts we had to make later on. It was not very expensive, and the budget allowed us to order extra. It saved us the headache of making a second order later in the semester, and avoided a second wait for the delivery.

Our scheduling at the beginning of the spring was very forgiving. We gave ourselves multiple weeks to finish parts that we expected to be done with in one. This was extremely helpful for when issues came up. When the waterjet was down, we had extra time to get the work done at an outside shop. This lenient time frame for fabrication allowed for our team to get our project done before the deadline we set for testing.

Overall our team learned a lot from this project as a whole. We would recommend that anybody taking on a similar project to learn from our mistakes and our successes.

8.0 Summary

The Ember Generator 2.0 was successfully designed and fabricated to be a portable system to help researchers further study wildfire and ember behavior to simulate wildfire conditions in a controlled environment. To simulate realistic wildfire conditions, the team generated a stainless steel combustion chamber and included an innovative shaking mechanism that was built from a redesigned massage gun to reduce cost and ensure consistent ember generation. The system included a straight pipe wind tunnel that was powered by a multispeed fan that was able to simulate wind speeds up to 16 ft/s.

Last semester we conducted engineering analysis like heat transfer calculations and projectile motion and conducted material selection to confirm the system would be able to withstand high temperatures and project embers at the required range of 3-5 feet. We went through various manufacturing challenges such as, combustion chamber fabrication issues, welding, waterjet cutting issues but we successfully got through those challenges. The team had to create innovative solutions to the problems like creating a separation flap from the initial burning area to the shaking mesh area which resulted in a more compact and safer final design. After finishing the system the team tested in the Cal Fire State facility in El Cajon and verified that the system was able to meet all the performance criteria including a consistent 10-minute burn time, portability, ember projection of at least 3-5 ft, and successfully used the embers to burn dry vegetation and simulated realistic wildfire conditions.

The project was presented at a very successful Design Day, where the system grabbed the attention of the audience and the results of our generator impressed those who stopped by. Throughout the process, the team learned about risk mitigation, manufacturing processes, design, innovative designs, and the importance of time management. The generator will now be used as a fire research tool for Dr. Luca Carmignani to further study ember behavior and support fire prevention efforts in Wildland-Urban Interface (WUI) areas.

9.0 References

1. Heat Founder. “90S Ignition Time Pit Boss Official Recommended Product Design Patented in the U.S. Silicon Nitride Hot Rod Igniter Pellet Grill Igniter Rod for Pitboss Camp Chef 100% Pit Boss Igniter Replacement HEATFOUNDER Hot Rod Igniter.” *Heat Founder - Make It Easy to Heat*, 2023, store.heatfounder.com/products/rapid-ignition-upgrade-kit-for-pellet-grills-smokers-quick-easy-installation?VariantsId=10097.
2. Marie, Anne. “Flame Temperatures Table for Different Fuels.” *ThoughtCo*, ThoughtCo, 2 May 2012, www.thoughtco.com/flame-temperatures-table-607307.
3. Suzuki, S., & Manzello, S. L. (2018). Characteristics of Firebrands Collected from Actual Urban Fires. *Fire Technology*, 54(6), 1533–1546. <https://doi.org/10.1007/s10694-018-0751-x>
4. Alec J. Petersen, Tirtha Banerjee; Characterizing firebrands and their kinematics during lofting. *Physics of Fluids* 1 October 2024; 36 (10): 106611. <https://doi.org/10.1063/5.0227024>

Appendix

URL for Google Drive:

https://drive.google.com/drive/folders/1DbLeajYMpVOXGXtHjeppBvR4VySrZKp?usp=drive_link

FDR from Fall 2025:

https://docs.google.com/document/d/1u8rwXahAsfq0XLBTUpnjANt04lIO70eqhMSu2_1OHn0/edit?usp=sharing

The image shows a "Signed Labor Hours Form" oriented vertically. The form contains a table with columns for "Date", "Time", "Activity", "Hours", and "Signature". The table is filled with handwritten data. At the bottom of the form, there are four signatures: "Mark M...", "Ryan Chase", "Sam Stoy", and "Paul Gray". The form is titled "Signed Labor Hours Form" at the bottom.

Signed Labor Hours Form

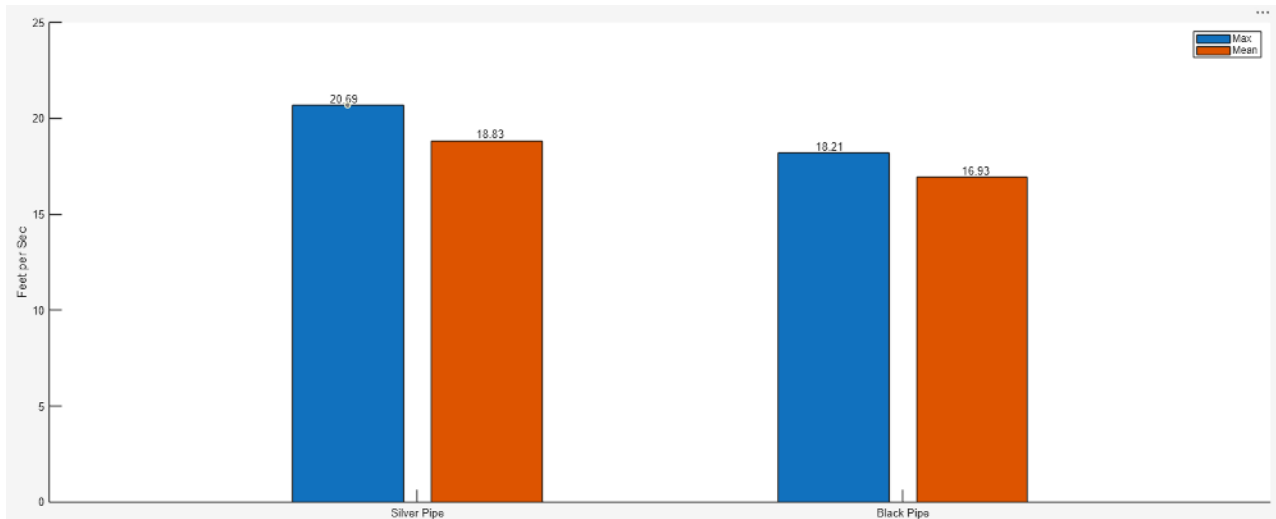


Figure A1. Wind Speed Test Results

- For the electrical schematics figures A2-A4, the goal of creating and designing them was to ultimately remove the battery that came with the massage gun with a direct DC power source. Mark Bruno, a Equipment/Systems Specialist of SDSU, helped design and create the circuit. Having this circuit also helps extend how far the circuit can be away from the massage gun, helping maintain a safe distance for the electronics away from the combustion chamber. The button switch in the middle of the circuit acts as a momentary break switch to turn off the massage gun when pressed. The circuit works as intended, removing both the battery replacing it with a DC barrel jack for power and the button turns off the massage gun when pressed. The DC power source was a universal adjustable source that was set to 7.5 volts.

- Software code listing (with descriptions)

```
#include <LiquidCrystal.h>
```

```
#include <math.h>
```

```
// LCD pins: RS, E, D4, D5, D6, D7
```

```
LiquidCrystal lcd(12, 11, 5, 4, 3, 2);
```

```
/*
```

```
* Hardware hookup
```

```
* Sensor      Arduino Pin
```

```
* Ground      Ground
```

```
* +10-12V     Vin
```

```
* Out         A0
```

```
* TMP         A2
```

```
* Needs 8v to work, 9 volt battery or barrel jack connection
```

```
*/
```

```

const int OutPin  = A0;
const int TempPin = A2;

void setup() {
  Serial.begin(9600);

  lcd.begin(16, 2);
  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print("Wind Sensor");
  delay(1000);
  lcd.clear();
}

void loop() {
  // read wind
  int windADunits = analogRead(OutPin);

  float windMPH = pow((((float)windADunits - 264.0) / 85.6814),
3.36814);

  // keep negative/invalid values from showing up
  if (windMPH < 0 || isnan(windMPH)) {
    windMPH = 0;
  }

  float windFPS = windMPH * 1.46667;

  // read temp
  int tempRawAD = analogRead(TempPin);
  float tempC = (((float)tempRawAD * 5.0) / 1024.0) - 0.400) /
0.0195;
  float tempF = (tempC * 9.0 / 5.0) + 32.0;

  // serial output
  Serial.print(windFPS);
  Serial.print(" Ft/s  ");
  Serial.print(tempF);
  Serial.println(" F");
}

```

```

// LCD output
lcd.clear();

lcd.setCursor(0, 0);
lcd.print("Wind:");
lcd.print(windFPS, 1);
lcd.print("ft/s");

lcd.setCursor(0, 1);
lcd.print("Temp:");
lcd.print(tempF, 1);
lcd.print("F");

delay(750);
}

```

Nozzle Diameter = 1/3ft

Cross-Sectional Area = $\pi \cdot 1/3^2/4 = 0.087ft^2$

CFM 100 = 100/60 = 1.667 CFS

$1.667/0.087 = 19.01ft/s^2$

$V_{ember} = 0.6 \cdot 19.01 = 11.459 ft/s$

$H_f = 3ft$

$t_f = \sqrt{\frac{(2 \cdot 36)}{386.1}} = 0.423s$

$R = 0.423s \cdot 11.459ft/s = 4.95ft$

Exit Speed: 100CFM

$f = 0.6$

Conduction Heat Rate: $- 112 * 80.56in^2 \frac{1800f^\circ - 77f^\circ}{7.47 * 10^{-2}} = 1.73 * 10^7 BTU/hr$

Radiation Heat Rate:

$0.75 * 1.714 * 10^{-9} (\frac{BTU}{hr} * ft^2 * R^4) * 80.56in^2 * (1477.59^4 K - 294.26^4 K) = 4.92 * 10^5 Btu/hr$

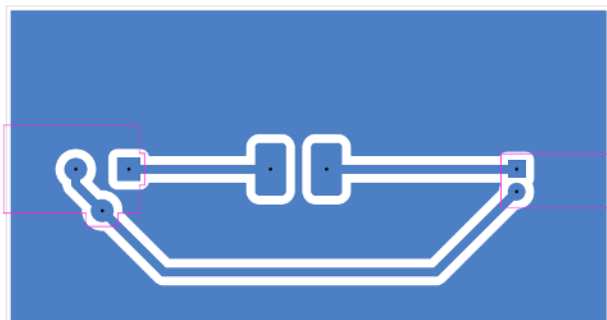


Figure A2. PCB Schematic for Massage Gun

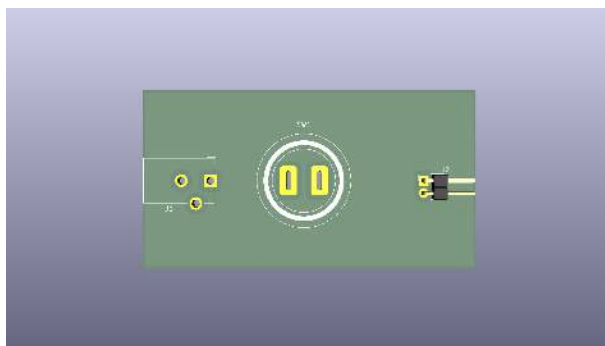


Figure A3. Projected Schematic for Circuit

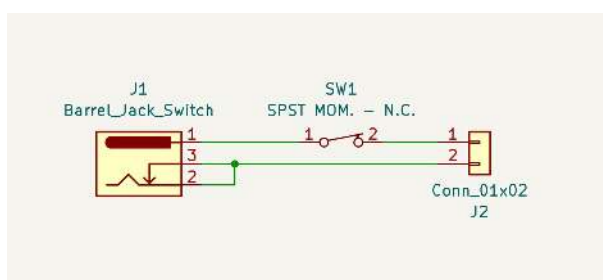


Figure A4. Electronic Circuit for Massage Gun